MATHEMATICAL PROOF PORTFOLIO

A Comprehensive Compilation of Discrete Structures, Induction, and Asymptotic Analysis

Problem 1: Matrix Workbench — Compatibility, Multiplication, and Complexity Reporter

An overview of fundamental statements and the optimal mathematical proof methodologies required to address them:

#	Mathematical Statement	Best Technique	Methodological Explanation
1	For every integer x, the product $x(x + 1)$ is even.	**Direct Proof**	The statement is an implication of the form "for all x, $P(x)$". We can directly show that the product is always divisible by 2 by considering that any two consecutive integers must include an even number.
2	For every positive integer n, $1 + 2 + 3 + \ldots + n = n(n + 1) / 2$	**Mathematical Induction**	This is a structural summation statement about all positive integers that features a clear base case ($n = 1$) and a natural inductive step that cleanly builds the sum for $k + 1$ from the sum for k.
3	If n^2 is even, then n is even.	**Proof by Contrapositive**	The logical contrapositive statement, "if n is odd, then n^2 is odd", is significantly less complex to prove directly using the algebraic definition of odd integers ($n = 2k + 1$).
4	Prove that $5n^2 + 3n + 20 = O(n^2)$.	**Big-O Formal Definition**	This requires a direct application of the formal asymptotic bounds definition, necessitating the derivation of specific witness constants c and n_0 such that the functional inequality holds for all $n \geq n_0$.
5	Disprove that for every positive integer n, $n! \leq n^2$.	**Disproof by Counterexample**	The universally quantified statement claims viability "for all n". Identifying a single concrete, valid index instance where the statement fails completely invalidates the universal claim.

Problem 2: Complete Direct Proof

Statement: If a and b are even integers, then $a + b$ is even.

DIRECT PROOF

Let a and b be arbitrary even integers.

By the foundational definition of an even integer, there must exist integers m and k such that:

$$a = 2m \quad \text{and} \quad b = 2k$$

We substitute these expressions into the sum $a + b$:

$$a + b = 2m + 2k = 2(m + k)$$

Because the set of integers is closed under addition, the term $(m + k)$ must be an integer. Let $q = m + k$ where q ∈ &Z;. Substituting back yields:

$$a + b = 2q$$

Therefore, by the definition of even numbers, $a + b$ is divisible by 2 and is explicitly an even integer.

Conclusion: If a and b are even integers, then $a + b$ is even. $\blacksquare$

Problem 3: Formal Asymptotic Proofs

1. Prove that $4n^2 + 7n + 25 = O(n^2)$

Claim: $4n^2 + 7n + 25 = O(n^2)$

Proof: We must find positive constants $c > 0$ and $n_0 > 0$ such that for all $n \geq n_0$:

$$4n^2 + 7n + 25 \leq c \cdot n^2$$

Assume an initial lower bound of $n \geq 1$. For all such values of n, it is structurally true that $7n \leq 7n^2$ and $25 \leq 25n^2$. Substituting these upper bounds into the expression yields:

$$4n^2 + 7n + 25 \leq 4n^2 + 7n^2 + 25n^2 = 36n^2$$

By selecting the explicit parameters $c = 36$ and $n_0 = 1$, the required inequality $4n^2 + 7n + 25 \leq 36n^2$ holds continuously for all $n \geq 1$.

Conclusion: Therefore, $4n^2 + 7n + 25 = O(n^2)$. ■

2. Prove that $5n^3 + 2n^2 + 10 = \Omega(n^3)$

Claim: $5n^3 + 2n^2 + 10 = \Omega(n^3)$

Proof: We must find positive constants $c > 0$ and $n_0 > 0$ such that for all $n \geq n_0$:

$$5n^3 + 2n^2 + 10 \geq c \cdot n^3$$

Assume a lower bound of $n \geq 1$. For all positive values of n, it is true that $2n^2 \geq 0$ and $10 \geq 0$. Dropping these strictly positive non-dominant terms yields the following valid inequality:

$$5n^3 + 2n^2 + 10 \geq 5n^3$$

By selecting the explicit parameters $c = 5$ and $n_0 = 1$, the inequality holds perfectly for all $n \geq 1$.

Conclusion: Therefore, $5n^3 + 2n^2 + 10 = \Omega(n^3)$. ■

3. Prove that $7n^2 + 3n + 100 = \Theta(n^2)$

Claim: $7n^2 + 3n + 100 = \Theta(n^2)$

Proof: To establish the tight bound Θ, we must prove both the upper bound (O) and lower bound (Ω).

Part A: Proof of the Upper Bound ($O(n^2)$)

For all $n \geq 1$, it is true that $3n \leq 3n^2$ and $100 \leq 100n^2$. Hence:

$$7n^2 + 3n + 100 \leq 7n^2 + 3n^2 + 100n^2 = 110n^2$$

This satisfies the definition of $O(n^2)$ with $c_1 = 110$ and $n_{0,1} = 1$.

Part B: Proof of the Lower Bound ($\Omega(n^2)$)

For all $n \geq 1$, since the linear and constant components are strictly additive and non-negative ($3n + 100 \geq 0$), we can drop them to formulate a lower constraint:

$$7n^2 + 3n + 100 \geq 7n^2$$

This satisfies the definition of $\Omega(n^2)$ with $c_2 = 7$ and $n_{0,2} = 1$.

Conclusion: By setting $c_2 = 7$, $c_1 = 110$, and $n_0 = \max(1, 1) = 1$, we confirm that $7n^2 \leq 7n^2 + 3n + 100 \leq 110n^2$ for all $n \geq 1$. Thus, the statement holds. ∎

4. Disprove the statement $n^2 + 10n = O(n)$

Proof by Contradiction:

Assume for contradiction that the statement is true: $n^2 + 10n = O(n)$. By definition, there must exist constant values $c > 0$ and $n_0 > 0$ such that for all $n \geq n_0$:

$$n^2 + 10n \leq c \cdot n$$

Since $n \geq n_0 > 0$, we can safely divide both sides of the inequality by the variable n:

$$n + 10 \leq c$$

This results in an absolute upper limit where the variable expression $n + 10$ is bounded above by a single fixed real number constant c. However, by the Archimedean property of real numbers, n can be made arbitrarily large, making $n + 10 > c$ for any value of $n > c - 10$. This directly contradicts our initial assumption.

Conclusion: The original statement is false; hence $n^2 + 10n \neq O(n)$. ■

Problem 4: Mathematical Induction Proof Portfolio

1. Sum of the First n Natural Numbers

Claim: For every positive integer n, $1 + 2 + \&dots; + n = n(n + 1) / 2$.

Base Case ($n = 1$):
Left-Hand Side (LHS): 1
Right-Hand Side (RHS): $1(1 + 1) / 2 = 2 / 2 = 1$. Since LHS = RHS, the base case is verified.

Inductive Hypothesis: Assume that the claim holds true for some arbitrary positive integer $n = k$:

$$1 + 2 + \&dots; + k = k(k + 1) / 2$$

Inductive Step: We must prove the statement holds for $n = k + 1$. We evaluate the sum up to $k + 1$:

$$1 + 2 + \&dots; + k + (k + 1) = [1 + 2 + \&dots; + k] + (k + 1)$$

Applying the Inductive Hypothesis to the bracketed expression:

$$= [k(k + 1) / 2] + (k + 1) = [k(k + 1) + 2(k + 1)] / 2$$

Factoring out the common binomial term $(k + 1)$:

$$= (k + 1)(k + 2) / 2$$

This matches the target form $(k + 1)((k + 1) + 1) / 2$ perfectly.

Conclusion: By mathematical induction, the formula holds for all positive integers n. $\blacksquare$

2. Sum of the First *n* Odd Positive Integers

Claim: For every positive integer n, $1 + 3 + 5 + \&dots; + (2n - 1) = n^2$.

Base Case ($n = 1$):
LHS: $2(1) - 1 = 1$; RHS: $1^2 = 1$. Since LHS = RHS, the base case holds.

Inductive Hypothesis: Assume the statement is true for $n = k$:

$$1 + 3 + 5 + \&dots; + (2k - 1) = k^2$$

Inductive Step: Prove the statement for $n = k + 1$. The sequence term at index $k + 1$ is $2(k + 1) - 1 = 2k + 1$. Summing up to this term:

$$1 + 3 + \&dots; + (2k - 1) + (2k + 1) = [1 + 3 + \&dots; + (2k - 1)] + (2k + 1)$$

Substituting the inductive hypothesis into the bracketed term:

$$= k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$$

Conclusion: By mathematical induction, the statement holds universally for all positive integers n. $\blacksquare$

3. Summation of Consecutive Products $n(n + 1)$

Claim: $1\cdot 2 + 2\cdot 3 + \&dots; + n(n + 1) = n(n + 1)(n + 2) / 3$.

Base Case ($n = 1$):
LHS: $1(1 + 1) = 2$; RHS: $1(1 + 1)(1 + 2) / 3 = (1 \cdot 2 \cdot 3) / 3 = 2$. The base case is correct.

Inductive Hypothesis: Assume the statement is true for $n = k$:

$$1\cdot 2 + 2\cdot 3 + \&dots; + k(k + 1) = k(k + 1)(k + 2) / 3$$

Inductive Step: Prove the formula for $n = k + 1$. Adding the next term in the sequence:

$$1\cdot 2 + \&dots; + k(k + 1) + (k + 1)(k + 2) = [1\cdot 2 + \&dots; + k(k + 1)] + (k + 1)(k + 2)$$

Apply the Inductive Hypothesis to resolve the sum block:

$$= [k(k + 1)(k + 2) / 3] + (k + 1)(k + 2)$$

Factor out the shared expressions $(k + 1)(k + 2)$ from both blocks:

$$= (k + 1)(k + 2) \cdot [k/3 + 1] = (k + 1)(k + 2) \cdot [(k + 3) / 3]$$

$$= (k + 1)(k + 2)(k + 3) / 3$$

This matches the target form $(k + 1)((k + 1) + 1)((k + 1) + 2) / 3$.

Conclusion: By mathematical induction, the formula holds for all positive integers n. ∎

4. Divisibility Property of $n^3 + 2n$

Claim: For every positive integer n, the expression $n^3 + 2n$ is divisible by 3 (denoted $3 \mid (n^3 + 2n)$).

Base Case ($n = 1$):
Evaluating the term: $1^3 + 2(1) = 3$. Since 3 is cleanly divisible by 3 ($3 = 3 \cdot 1$), the base case is true.

Inductive Hypothesis: Assume the statement is true for $n = k$, meaning there exists an integer m such that:

$$k^3 + 2k = 3m$$

Inductive Step: We evaluate the expression for the index $n = k + 1$:

$$(k + 1)^3 + 2(k + 1) = (k^3 + 3k^2 + 3k + 1) + (2k + 2)$$

Regrouping the variables to match the inductive hypothesis structure:

$$= (k^3 + 2k) + 3k^2 + 3k + 3$$

Substituting the inductive hypothesis value ($3m$) into the equation:

$$= 3m + 3(k^2 + k + 1) = 3(m + k^2 + k + 1)$$

Since m, k are integers, the evaluated term $(m + k^2 + k + 1)$ must also be an integer. Thus, the total expression is a multiple of 3.

Conclusion: By mathematical induction, $3 \mid (n^3 + 2n)$ holds for all positive integers n. ∎

5. Correctness Proof for Summation Loop Invariant Algorithm

Loop Invariant: At the start of the k-th iteration of the execution loop, the working tracking variable sum holds the accumulated total of the first k elements: $sum = 1 + 2 + \ldots + k$.

Initialization (Base Case $k = 0$): Prior to the first iteration of the loop, the variable sum is initialized to 0. This accurately matches the empty sum definition, confirming the invariant before loop execution begins.

Maintenance (Inductive Step): Assume the invariant holds true at the start of loop iteration k, meaning $sum = 1 + 2 + \ldots + k$. During the loop execution step, the control logic increments the structural pointer and adds the value $k + 1$ to the accumulator:

$$sum_{new} = sum_{old} + (k + 1) = (1 + 2 + \ldots + k) + (k + 1)$$

This re-establishes the invariant for the next state, confirming maintenance across iterations.

Termination: The loop terminates precisely after executing n iterations. Upon termination, the invariant guarantees that $sum = 1 + 2 + \ldots + n$. The algorithm then outputs this value, which matches the closed-form calculation $n(n + 1) / 2$ derived in our previous structural induction proofs. ∎

Problem 5: Proof by Contradiction, Contrapositive, and Counterexample

1. Infinitely Many Prime Numbers

Proof by Contradiction:

Assume for contradiction that the set of all prime numbers is finite. Let this complete set of primes be listed as:

$$P = \{p_1, p_2, \ldots, p_n\}$$

We construct a new integer N by multiplying all known prime numbers together and adding 1:

$$N = (p_1 \cdot p_2 \cdot \ldots \cdot p_n) + 1$$

Since $N > 1$, by the Fundamental Theorem of Arithmetic, N must be divisible by at least one prime number p_i from the universe of numbers. However, if any prime p_i from our set divides N, it must also divide the difference between N and the product of the primes:

$$N - (p_1 \cdot p_2 \cdot \ldots \cdot p_n) = 1$$

This implies that the prime number p_i divides 1, which is impossible since all prime numbers are strictly greater than 1 ($p_i > 1$). Thus, N must possess a prime factor that is not included in our assumed finite set P.

This directly contradicts the assumption that P contains every prime number. Therefore, the collection of prime numbers must be infinite. ■

2. Non-existence of a Smallest Positive Rational Number

Proof by Contradiction:

Assume for contradiction that there exists a smallest positive rational number, which we denote as r. By this definition, $r > 0$ and $r \in$ ***&Q;***.

Now, consider the alternative value $x = r / 2$. We analyze the properties of x:

- Since $r > 0$, dividing by 2 preserves the sign, meaning $x = r / 2 > 0$.
- Because r is rational, it can be written as a/b for integers a, b. Thus, $x = a / (2b)$, which is a ratio of integers and is also rational ($x \in$ ***&Q;***).
- Since $r > 0$, it is structurally true that $r / 2 < r$, meaning $x < r$.

This identifies x as a valid positive rational number that is strictly smaller than r, which directly contradicts the assumption that r is the smallest positive rational number.

Conclusion: There is no smallest positive rational number. $\blacksquare$

3. Implication: If n^2 is Odd, then n is Odd

Proof by Contrapositive:

The original statement is of the form P ***&implies;*** Q: "If n^2 is odd (P), then n is odd (Q)."

The logical contrapositive is structurally defined as $\neg Q$ ***&implies;*** $\neg P$: "If n is not odd (meaning n is even), then n^2 is not odd (meaning n^2 is even)."

We proceed with a direct proof of this contrapositive statement. Let n be an even integer. By definition, there exists an integer k such that:

$$n = 2k$$

We evaluate the square of this integer:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

Let $m = 2k^2$. Since k is an integer, m must also be an integer. Substituting this back yields $n^2 = 2m$, which satisfies the definition of an even integer.

Because the contrapositive statement $\neg Q$ ***&implies;*** $\neg P$ has been proved true, the logically equivalent original statement P ***&implies;*** Q is also true. ■

4. Universal Disproof via Counterexample

Statement to Disprove: For every integer n, if n^2 is positive, then n is positive.

Disproof by Counterexample:

To disprove this universally quantified statement, we must find a single instance where the premise is true but the conclusion is false.

Let us choose the concrete integer value:

$$n = -1$$

We evaluate the two conditions for this value:

- Premise Check: $n^2 = (-1)^2 = 1$. Since $1 > 0$, the square is positive. The premise holds true.
- Conclusion Check: $n = -1$. Since $-1 \leq 0$, the number is not positive. The conclusion is false.

This single counterexample demonstrates an integer whose square is positive but the number itself is not positive. Therefore, the universal statement is false. ■